Ecology: abundance and distribution

9C Threat from disease · Lesson 6 of 7

Date _____ Class _____

Today I can 今天我能

- ☐ I can choose suitable apparatus for measuring the abundance and distribution of organisms.
- ☐ I can use data from a sample to estimate a population size and say how accurate the estimate is.

What happens today

1. Do Now: counting the uncountable
2. Learn: Abundance and sampling · Estimating population size · and more
3. Practical: Quadrat sampling
4. Challenge, then the exit ticket

Student Book pages 48 to 49 · Workbook pages 34 to 35

What I learned today (one or two sentences) 今天我学到了：

Key Vocabulary

Term	中文	Definition
abundance	丰度	how common a species is in a habitat
distribution	分布	how organisms are spread out in an area
population	种群	all the organisms of one species in an area
sample	样本	a small selected part of the total number
estimate	估算	an approximate calculation of something
quadrat	样方	a square frame used to sample plants or slow-moving animals
transect	样线	a line along which samples are taken
pitfall trap	陷阱	a container sunk into the ground to catch small ground animals
sweep net	捕虫网	a net swept through long grass to catch insects
random	随机的	chosen by chance, not by the person
biodiversity	生物多样性	the number of different species in an area

Command words 指令词: Calculate 计算 · Complete 完成（填写） · Describe 描述 · Draw 画出 · Estimate 估算 · Give 给出 · Match 连线（配对） · Predict 预测 · Suggest 提出（建议） · Tick 打勾

Quick Review

counting the uncountable

During a disease outbreak, scientists cannot count every infected person, so they estimate.

1 What does 'estimate' mean?

2 Suggest how you could estimate the number of dandelions on the school field without counting them all.

3 Name one abiotic and one biotic factor that could affect how many dandelions there are.

Core 1 of 6

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

1 Match each word to its definition. Draw a line.

abundance	an approximate calculation of something
distribution	the number of different species in an area
sample	how common a species is in a habitat
estimate	a small selected part of the total number
biodiversity	how organisms are spread out in an area

Core 2 of 6

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

2 Tick one box in each row to show the best method for sampling each organism.

Organism	pitfall trap	sweep net	quadrat	pooter
ground beetles				
insects in long grass				
daisies on a lawn				
tiny insects on a leaf				
limpets on a rocky shore				

Core 3 of 6

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

3 Students counted one species of snail in eight quadrats placed at random in a field 30 m by 25 m. Each quadrat was 0.5 m by 0.5 m.

Quadrat	1	2	3	4	5	6	7	8
Number of snails	3	1	0	2	4	1	2	3

a) Calculate the total number of snails counted.

b) Calculate the total area sampled by the eight quadrats, in m2.

Core 4 of 6

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

3 *continued*

c) Calculate the area of the whole field, in m2.

d) Estimate the population of this snail species in the whole field. Show your working.

Core 5 of 6

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

4 Record your own quadrat results, then estimate the population of your plant in the 10 m by 10 m area.

a) Complete the table.

Quadrat	Coordinates	Number of plants
1		
2		
3		
4		
5		
6		
7		
8		
total		

Core 6 of 6

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

4 *continued*

b) Estimate the population: total x 100 m2 / 2 m2. Show your working.

Challenge 1 of 3

When Core is finished. Full sentences. 完成基础题后做；请写完整句子。

C1 Ecologists compared the biodiversity of birds in rainforest and in oil palm plantations that replaced some of the forest. They walked along transects through each area, recording every bird they saw or heard.

a) Suggest why the scientists used transects rather than quadrats.

b) Give two things the scientists should keep the same so that the results from the two areas can be compared fairly.

Challenge 2 of 3

When Core is finished. Full sentences. 完成基础题后做；请写完整句子。

C1 *continued*

c) Oil palm plantations have trees planted in rows, all the same age; rainforest has many kinds of trees of different sizes. Predict in which area the birds would show a more regular distribution, and give a reason.

Challenge 3 of 3

When Core is finished. Full sentences. 完成基础题后做；请写完整句子。

C2 There are many types of grass, each with slightly different leaf shapes. Describe how you would find out which of four species was the most abundant in a large field.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 Why do ecologists take samples instead of counting every organism?

2 5 daisies were found in one 0.25 m2 quadrat. The lawn is 40 m2. Estimate the daisy population.

3 Give one reason why quadrats should be placed at random.

Today I can: Goal 1 ☐ Goal 2 ☐